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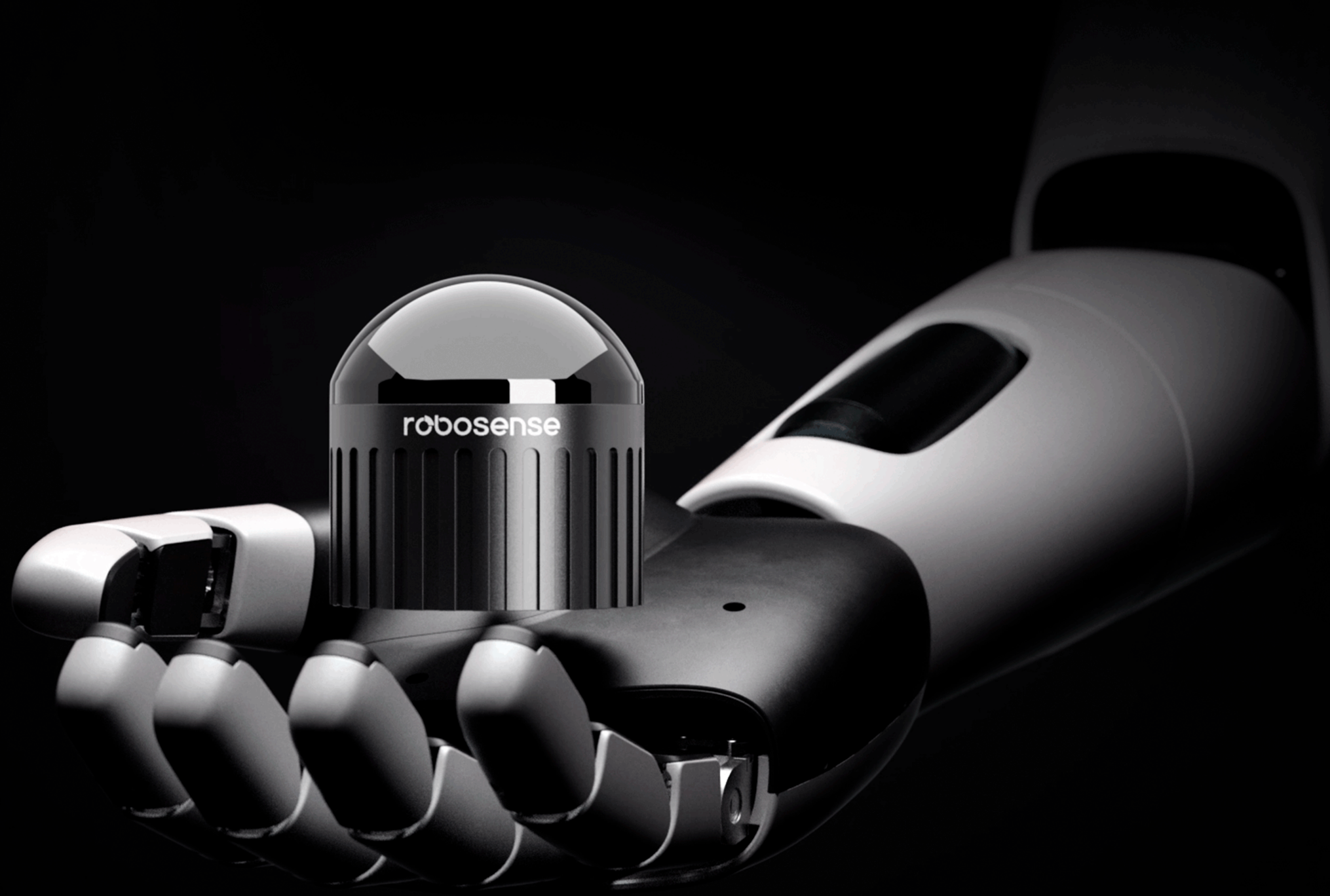

Airy

The World's First 192-Beam Hemispherical Digital LiDAR

RoboSense / Suteng Innovation Technology Co., Ltd.

RoboSense Global Headquarters - Building 9, Block 2, Zhongguan Honghualing Industry Southern District, 1213 Liuxian Avenue, Taoyuan Street, Nanshan District, Shenzhen, China / 0755-86325830 / service@robosense.cn

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Overview

Airy is the first digital hemispherical LiDAR on the R platform, featuring a chip-based transceiver solution and a high-sensitivity digital detection scheme. Compact in size, comparable to a ping-pong ball, it delivers an ultra-wide hemispherical FOV with 360° horizontal and 90° vertical coverage, encompassing a 60-meter radius range. With 1.72 million points per second and ±1cm detection accuracy, Airy provides real-time detection of object size, contour, and distance. Its well-structured, algorithm-friendly point clouds enable exceptional performance in obstacle avoidance, mapping, and navigation tasks.

Airy also integrates an IMU, supports a wide variety of robots operating seamlessly in diverse lighting conditions, enabling all-weather continuous operations and advancing the intelligent application of robots across all scenarios.

Product Specifications

Range	30m@10%	#of Lines	192/96/48
Maximum range	60m	Range Precision	1cm@1σ
FOV (H×V)	360°×90°	Blind spot	＜0.1m
Angular resolution (H×V)	Avg.0.4°x0.47°	Laser safety	Class 1 eye safety
Frame rate	10Hz	Power consumption	<8W
Points per second (single return mode)	~1,720,000pts/s (192-Beam)) ~860,000pts/s (96-Beam) ~430,000pts/s (48-Beam)	Points per second (Dual return mode)	~3,440,000pts/s (192-Beam) ~1,720,000pts/s (96-Beam) ~860,000pts/s (48-Beam)
Ethernet connection	100Base-TX	Output	UDP packets over Ethernet
Dimension*	ϕ60mm×H63mm	UDP packet include	Spatial Coordinates Intensity Timestamp, etc.
Weight (without cabling)	＜240g	Operating voltage	9V ~ 32V
Operating temperature	-40°C ~ +60°C	Storage temperature	-40°C ~ +85°C
Time synchronization	$GPRMC with 1PPS, PTP&gPTP	Ingress protection	IP67、IP6K9K

*Product specifications may vary by version. For specific information, please refer to the structural drawings or the 3D digital model.

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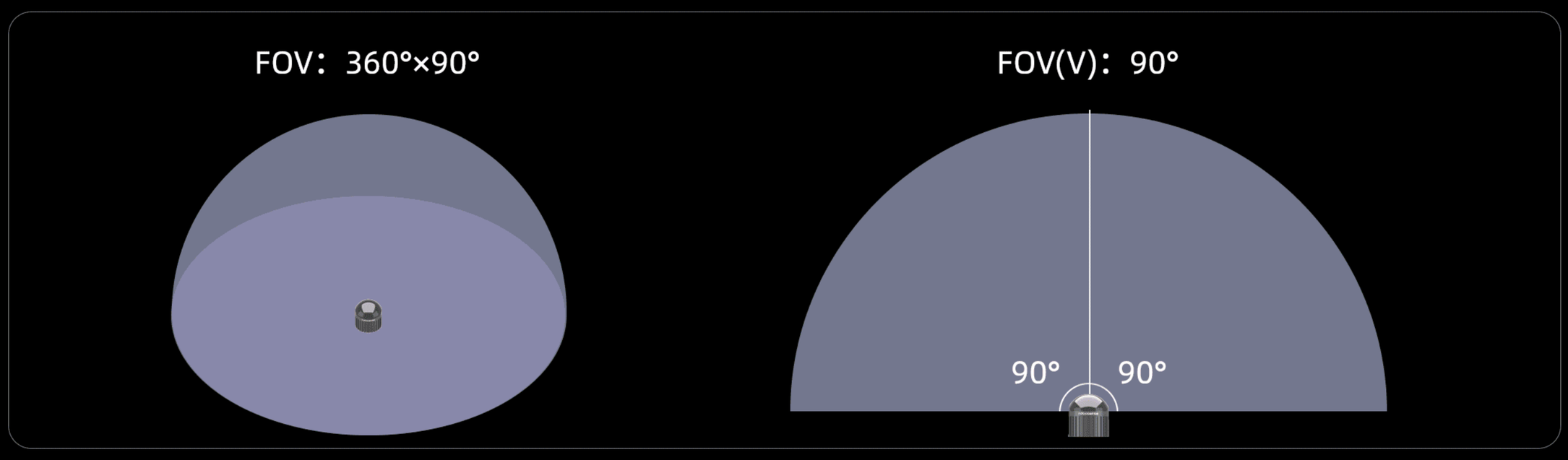
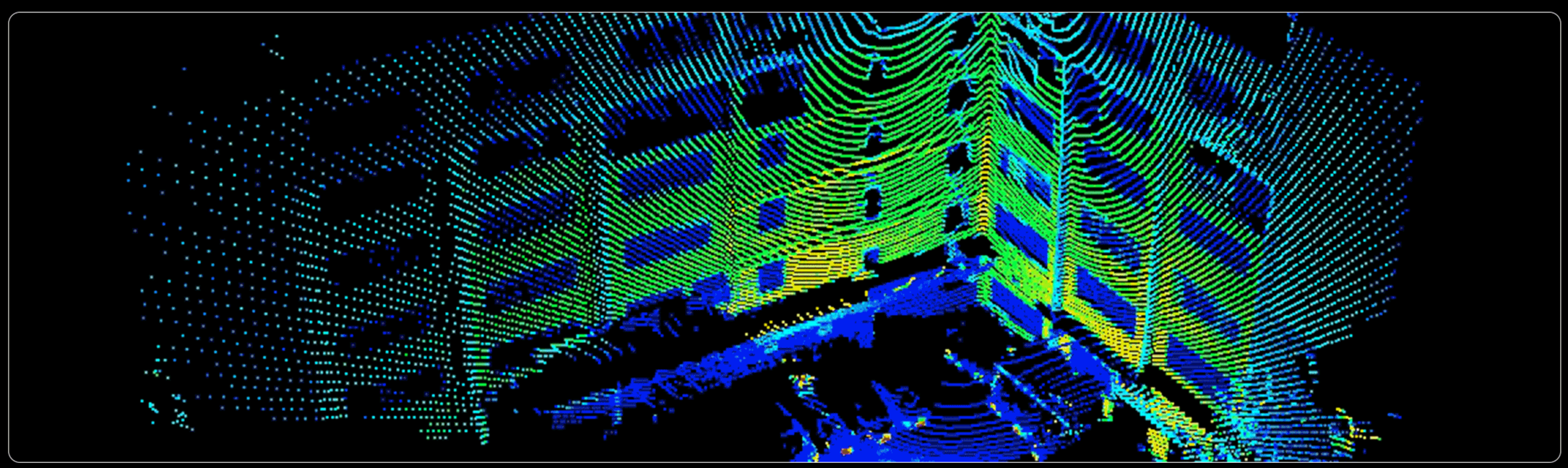
Perfect Hemispherical FOV

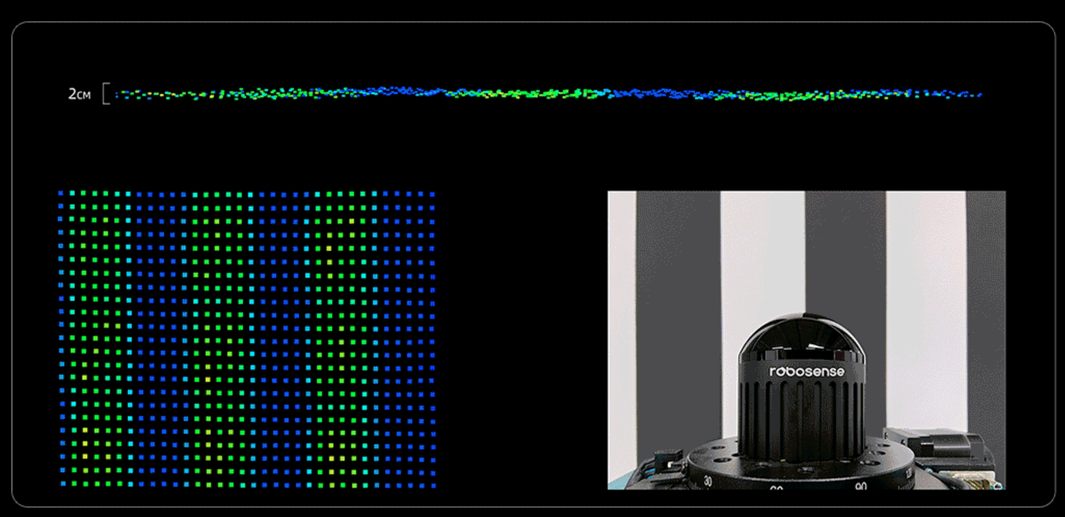


Neat and High Resolution Point Cloud

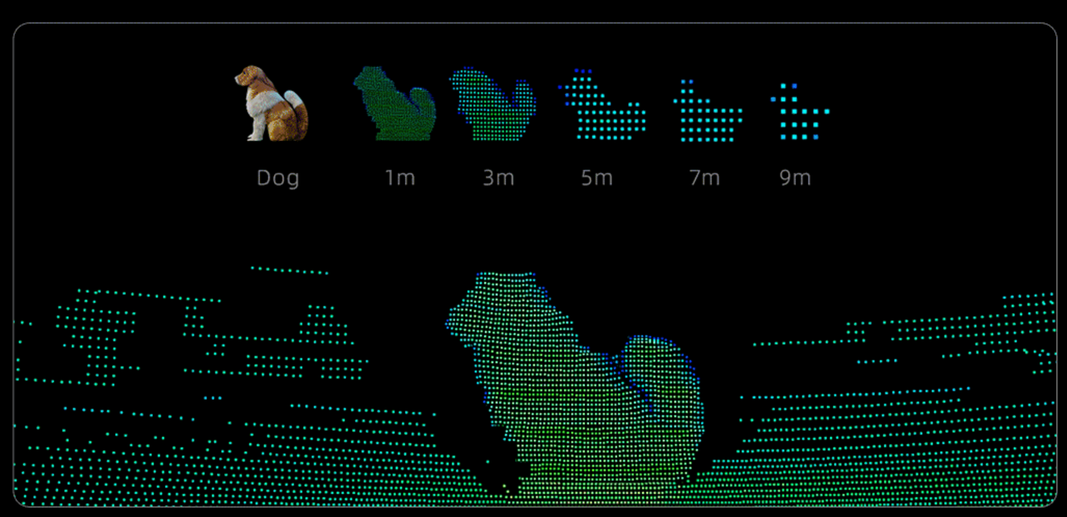
Ultra-wide FOV



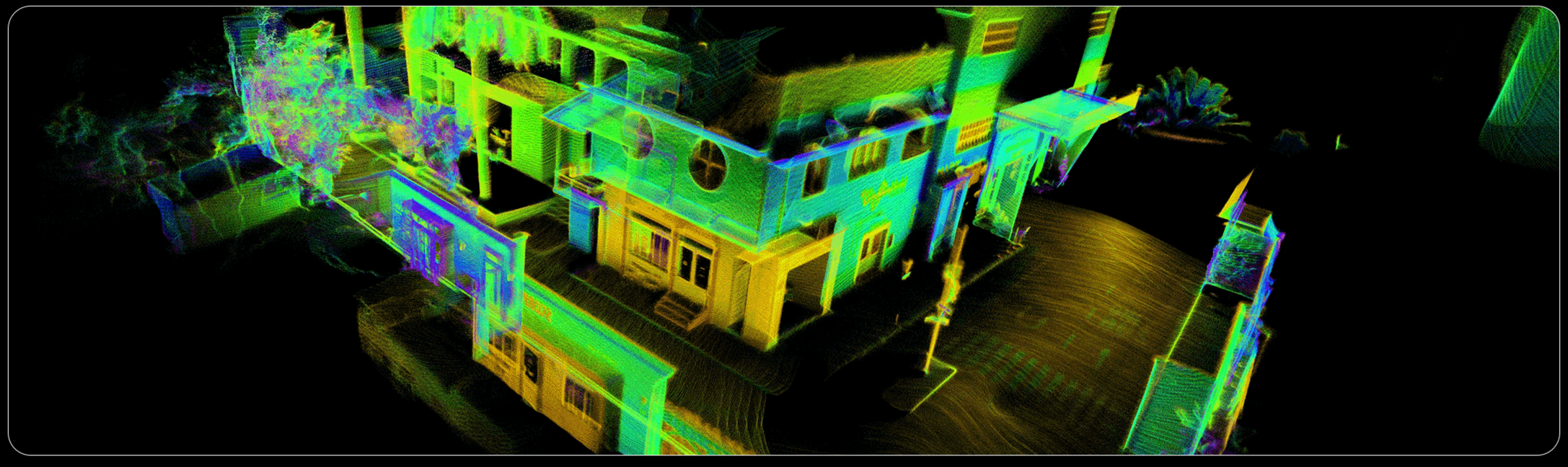
Achieves a distance measurement accuracy of up to ±1cm



High-Definition Detection of Small Animals



3D SLAM



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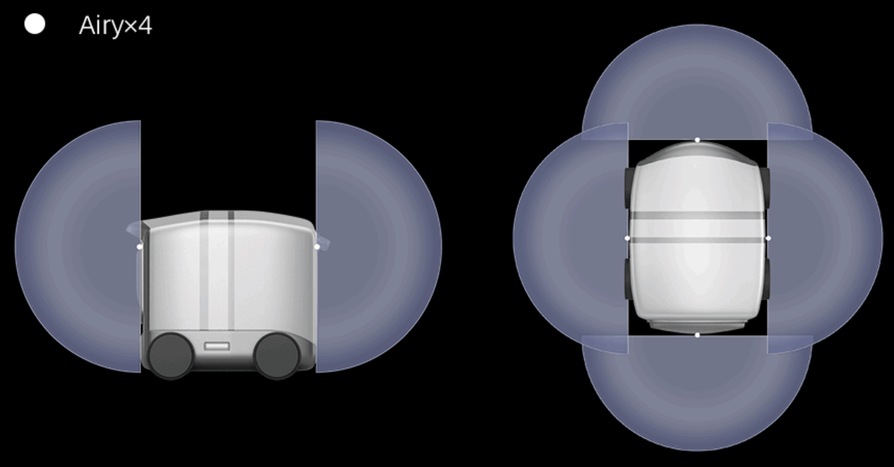
Applications

Airy delivers an ultra-wide hemispherical FOV with 360° horizontal and 90° vertical coverage, spanning a 120-meter diameter range. This significantly enhances robots' efficiency in obstacle avoidance, mapping, and localization.

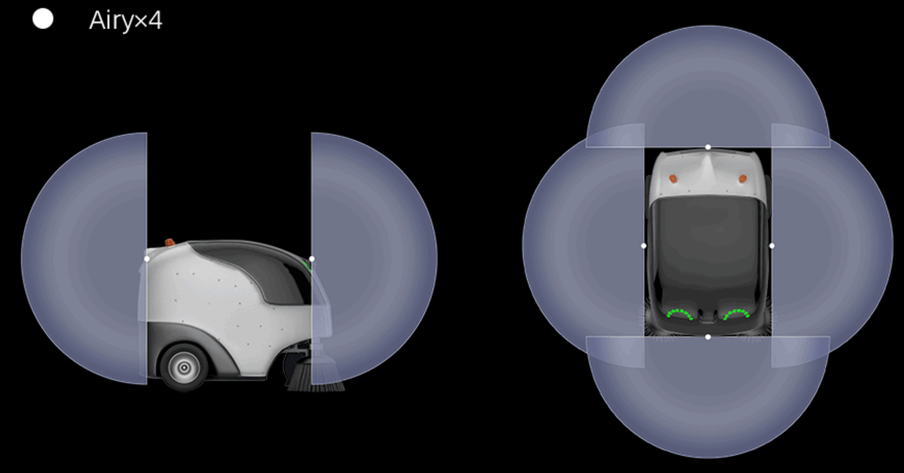
Autonomous Delivery Robot

- Airy×4



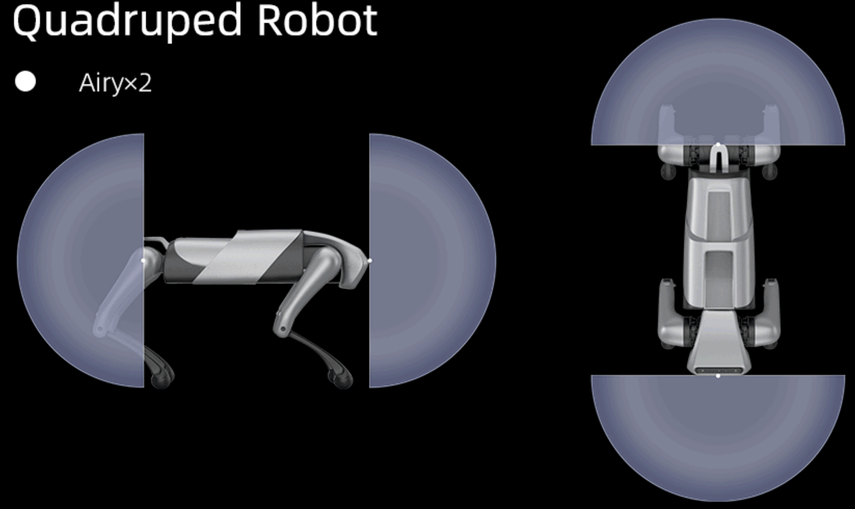
Cleaning Robot

- Airy×4



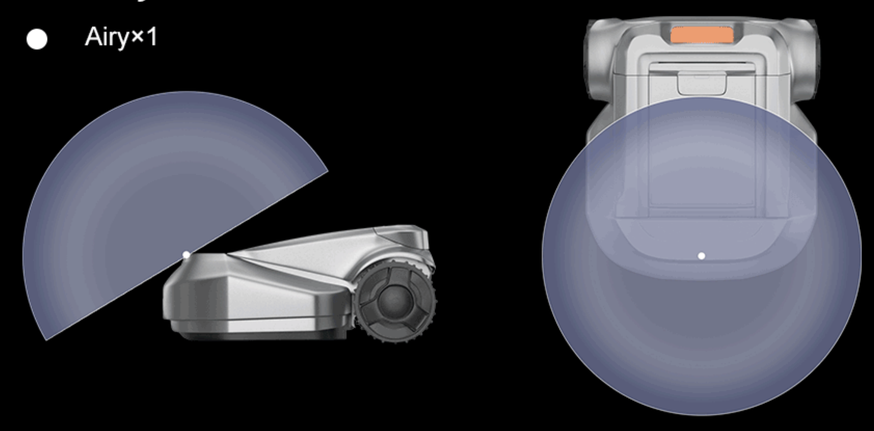
Quadruped Robot

- Airy×2



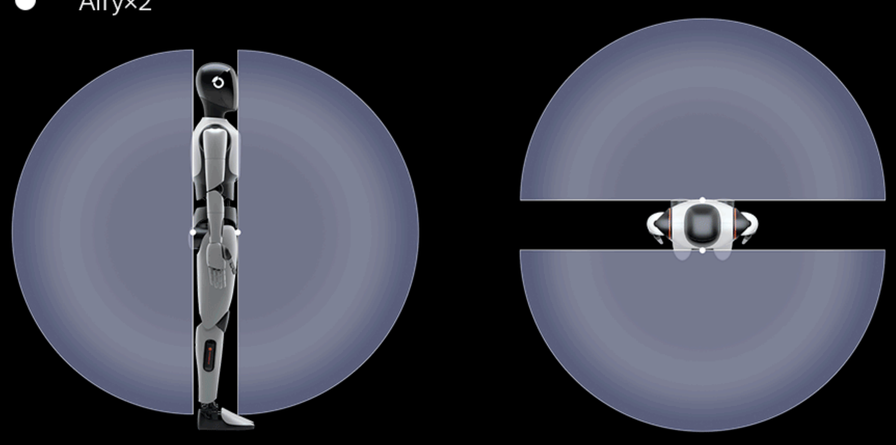
Courtyard Robot

- Airy×1



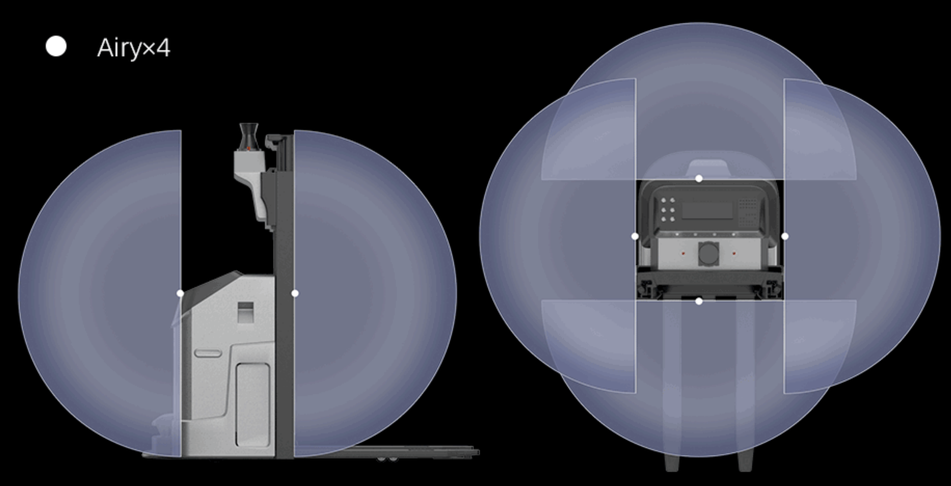
Humanoid Robot

- Airy×2



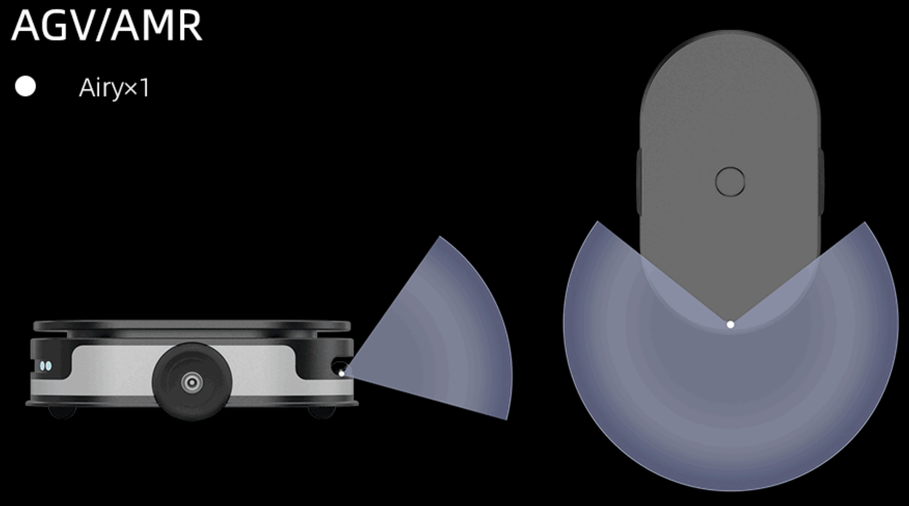
Unmanned Forklift

- Airy×4



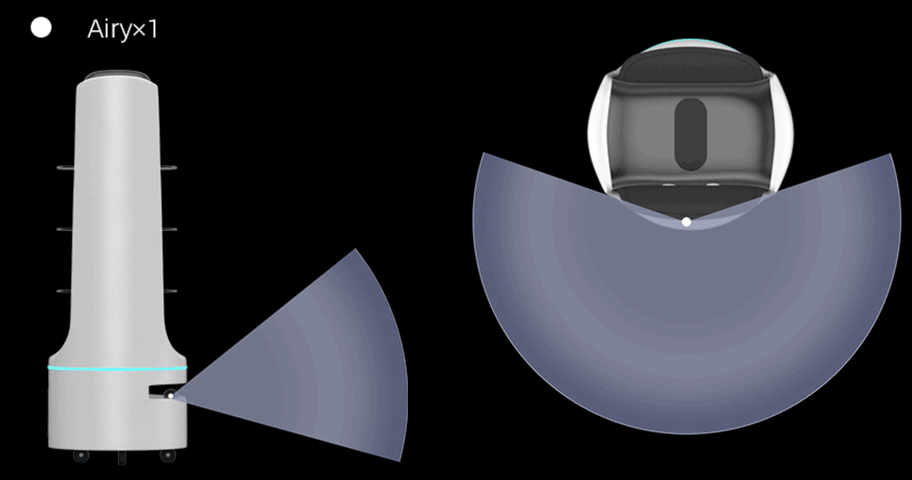
AGV/AMR

- Airy×1



Service Robot

- Airy×1



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